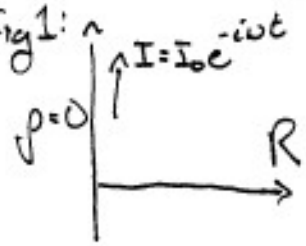
Discussion Problem #

In this problem we are considering the case of an infinitely long current running along the $\hat{z}$-axis which has a time dependence of $I(t) = I_0 e^{-i\omega t}$. The goal is to find the $\vec{E}$ and $\vec{B}$ fields in both near and far-field limits at a radius R.

Fig 1:



We can find $\vec{E}$ and $\vec{B}$ by first solving for the vector potential $\vec{A}(t, \vec{R})$. In the integral form we can write $\vec{A}$ as,

$$\vec{A}(t, \vec{R}) = \frac{\mu_0}{4\pi} \int d^3r \, \frac{\vec{j}\left(t - \frac{|\vec{R} - \vec{r}|}{c}, \vec{r}\right)}{|\vec{R} - \vec{r}|}$$

Since the current exists only at $\rho = 0$ we can write the current density as,

$$\vec{j}\left(t - \frac{|\vec{R} - \vec{r}|}{c}\right) = \vec{I}\left(t - \frac{|\vec{R} - \vec{r}|}{c}\right) \delta^2(\rho) = I_0 e^{-i\omega t} e^{i\omega \frac{|\vec{R} - \vec{r}|}{c}} \delta^2(\rho) \, \hat{z}$$

To understand $|\vec{R} - \vec{r}|$ we use the following convention (Fig. 2) considering an infinitesmal line element of charge dz at a height z above our origin which is set for convience at the same z-position as the observation point. Thus we see,

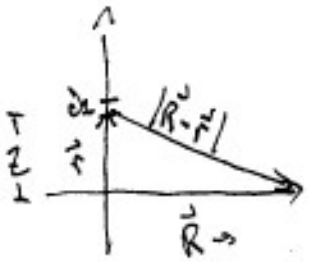


Fig 2

$$|\vec{R} - \vec{r}| = \sqrt{R^2 + z^2}$$

So, $$\vec{A}(t, \vec{R}) = \frac{\mu_0}{4\pi} \int d^3r \, \frac{I_0 e^{-i\omega t} e^{i\omega \frac{\sqrt{z^2 + R^2}}{c}}}{\sqrt{z^2 + R^2}} \delta^2(\rho)$$

$$\vec{A}_z(t,\vec{R}) = \frac{\mu_0 I_0 e^{-i\omega t}}{4\pi}\int d^3r\, \frac{e^{i\omega\frac{\sqrt{z^2+R^2}}{c}}}{\sqrt{z^2+R^2}}\,\delta^2(\rho)$$

We can rewrite the volume element as

$$d^3r = d^2\rho\, dz$$

~~which comes from the cylindrical symmetry where $d^2\rho$ is a circular "area" slice w/ some height dz.~~

$$\vec{A}_z(t,\vec{R}) = \frac{\mu_0 I_0 e^{-i\omega t}}{4\pi}\int_{-\infty}^{\infty} dz \int_0^{R} d^2\rho\, \frac{e^{i\omega\frac{\sqrt{z^2+R^2}}{c}}}{\sqrt{z^2+R^2}}\,\delta^2(\rho)$$

$$= \frac{\mu_0 I_0 e^{-i\omega t}}{4\pi}\int_{-\infty}^{\infty} dz\, \frac{e^{i\omega\sqrt{z^2+R^2}/c}}{\sqrt{z^2+R^2}}$$

Now we consider the two length scales in the problem

$\lambda \to$ wavelength of emitted radiation

$R \to$ distance to the observation point.

Hence, the far-field limit corresponds to $\frac{R}{\lambda} \gg 1$, and the near-field limit $\to \frac{R}{\lambda} \ll 1$

Re-writing $\lambda \sim \frac{1}{k} \sim \frac{c}{\omega}$ we can see that, $\frac{R}{\lambda} \sim \frac{\omega R}{c}$

so,

Near-Field Limit $\frac{\omega R}{c} \ll 1$

Far-Field Limit $\frac{\omega R}{c} \gg 1$

From this point forward we will let $\frac{\omega R}{c} = \chi$

So,

$$\vec{A}_z(t,\vec{R}) = \frac{\mu_0 I_0 e^{-i\omega t}}{4\pi}\int_{-\infty}^{\infty} dz\, \frac{e^{i\omega\frac{R}{c}\sqrt{1+\left(\frac{z}{R}\right)^2}}}{R\left(1+\left(\frac{z}{R}\right)^2\right)^{\frac{1}{2}}} \qquad \text{letting } u = \frac{z}{R}, \quad du = dz/R$$

$$= \frac{\mu_0 I_0 e^{-i\omega t}}{4\pi}\int_{-\infty}^{\infty} \frac{R\,du\, e^{i\chi\sqrt{1+u^2}}}{R\sqrt{1+u^2}} = \frac{\mu_0 I_0 e^{-i\omega t}}{2\pi}\int_0^{\infty} du\, \frac{e^{i\chi\sqrt{1+u^2}}}{\sqrt{1+u^2}}$$

Our problem is now to evaluate the dimensionless integral $\int_0^{\infty} du\, \frac{e^{i\chi\sqrt{1+u^2}}}{\sqrt{1+u^2}}$ in the near/far fields.

1.) In the far-field, $\chi \gg 1$.

$\vec{A}_z \sim \int_0^{\infty} du\, \frac{e^{i\chi\sqrt{1+u^2}}}{\sqrt{1+u^2}}$ The $e^{i\chi\sqrt{1+u^2}}$ will oscillate quickly for $\chi \gg 1$ but as u grows it will begin to oscillate even more rapidly. For these large values of u the integral over this region $\to 0$ thus, only small values of u will apply.

$(1+u^2)^{\frac{1}{2}} \simeq 1 + \frac{1}{2}u^2$

Since the integral will depend more on $e^{i\chi}$ the exponential then the denominator term we can drop powers of u^2 in the denominator.

$$\vec{A}_z \sim \int_0^{\infty} du\, \frac{e^{i\chi\sqrt{1+u^2}}}{\sqrt{1+u^2}} \simeq \int_0^{\infty} du\, \frac{e^{i\chi(1+u^2/2)}}{1} = e^{i\chi}\int_0^{\infty} du\, e^{i\chi\frac{u^2}{2}} = \frac{e^{i\chi}}{2}\sqrt{\frac{2\pi}{i\chi}}$$

Recovering the coeff,

$$\vec{A}_z(t,\vec{R}) = \frac{\mu_0 I_0 e^{-i\omega t}}{2\pi}\,\frac{e^{i\chi}}{2}\sqrt{\frac{2\pi}{i\chi}} = \frac{\mu_0 I_0}{4\pi} e^{-i\omega t + i\omega R/c}\sqrt{\frac{2\pi c}{i\omega R}} = \frac{\mu_0 I_0}{4\pi} e^{i\omega\left(\frac{R}{c}-t\right)}\sqrt{\frac{2\pi c}{i\omega R}}$$

2.) Near-field approx. $x \ll 1$

$$\vec{A}_z \sim \int_0^\infty \frac{e^{ix\sqrt{1+u^2}}}{\sqrt{1+u^2}}\,du$$

Since $x \ll 1$ then only large values of u will be important. Thus,

$$\vec{A}_z \simeq \int_0^\infty \frac{e^{ix\sqrt{u^2}}}{\sqrt{u^2}}\,du = \int_0^\infty du\,\frac{e^{ixu}}{u}$$

This integral diverges at $u \to 0$ but since we are only considering large u we can ~~ac~~ choose our lower limit to be $\to 1$. At extremely large u it begins to oscillate rapidly and thus any integral in that range $\to 0$ so we can choose our upper limit to be some large number where here we use $1/x$ since $x \ll 1$. In this range the exponential can be ignored.

$$\sim \int_0^\infty du\,\frac{e^{ixu}}{u} \simeq \int_1^{1/x} \frac{du}{u} = \ln\left(\tfrac{1}{x}\right) - \ln(1) = \ln\left(\tfrac{1}{x}\right) = -\ln\left(\tfrac{\omega R}{c}\right)$$

Thus,

$$\vec{A}_z \simeq \frac{\mu_0 I_0}{2\pi} e^{-i\omega t} \cdot \left[-\ln\left(\tfrac{\omega R}{c}\right)\right]$$

$$\vec{B} = \vec{\nabla} \times \vec{A} = -\frac{\partial A_z}{\partial R}\hat{\phi} \Rightarrow B_\phi = \frac{\mu_0 I_0 e^{-i\omega t}}{2\pi}\frac{\partial}{\partial R}\left(\ln\left(\tfrac{\omega R}{c}\right)\right) = \frac{\mu_0 I_0 e^{-i\omega t}}{2\pi}\frac{1}{R}$$

For the far-field case $x \gg 1$

$$A_z \sim \frac{\mu_0 I_0}{4\pi} e^{-i\omega t + i\omega \frac{R}{c}} \sqrt{\frac{2\pi c}{i\omega R}} = \frac{\mu_0 I_0 e^{-i\omega t}}{4\pi} e^{ix} \sqrt{\frac{2\pi c}{i\omega R}}$$

$$\cancel{\vec{B} = \vec{\nabla} \times \vec{A} = -\frac{\partial A_z}{\partial R} \hat{\phi} \Rightarrow B_\phi = -\frac{\mu_0 I_0 e^{-i\omega t}}{4\pi} \frac{2}{i} \sqrt{\frac{2\pi c}{i\omega}} \frac{\partial}{\partial R}\left(e^{i\omega \frac{R}{c}} \sqrt{\ }\right.}$$

$$\vec{B} = \vec{\nabla} \times \vec{A} = -\frac{\partial A_z}{\partial R} \hat{\phi} \qquad A_z = \frac{\mu_0 I_0}{4\pi} e^{-i\omega t} \sqrt{\frac{2\pi c}{i\omega}} e^{i\omega \frac{R}{c}} / \sqrt{R}$$

$$B_\phi = -\frac{\mu_0 I_0 e^{-i\omega t}}{4\pi} \sqrt{\frac{2\pi c}{i\omega}} \underbrace{\frac{\partial}{\partial R}\left(e^{i\omega R/c} \frac{1}{\sqrt{R}}\right)}_{-\frac{e^{i\omega \frac{R}{c}}}{2R^{3/2}} + \frac{i\omega}{c\sqrt{R}} e^{i\omega \frac{R}{c}}}$$

In the far-field limit ($\frac{\omega R}{c} \gg 1 \Rightarrow \frac{\omega}{c} \gg R$) we see that the second term will contribute more ~~for the~~ thus, we can ignore the first term.

$$B_\phi = -\frac{\mu_0 I_0 e^{-i\omega t}}{4\pi} \sqrt{\frac{2\pi c}{i\omega}} \frac{i\omega}{c} \frac{1}{\sqrt{R}} e^{i\omega R/c} = -\frac{\mu_0 I_0 e^{-i\omega t}}{4\pi} \sqrt{\frac{2\pi(i\omega)}{cR}} e^{i\frac{\omega R}{c}}$$